

Lineare Algebra I

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October 25, 2025

1 Introduction

Lec 1

1.1 Fibonacci Sequences

Definition 1.1:

The fibonacci sequence is defined as follows:

$$a_n = \begin{cases} 0, & n = 0 \\ 1, & n = 1 \\ a_{n-1} + a_{n-2}, & n \geq 2 \end{cases}$$

The first few terms of the sequence are: 0, 1, 1, 2, 3, 5, 8, 13, ...

Today we try to find an explicit formula for a_n .

From high school we know both the arithmetic and geometric sequences:

Definition 1.2: Arithmetic Sequence

An arithmetic sequence is defined as follows:

$$a_n = \begin{cases} a, & n = 0 \\ a_{n-1} + d, & n \geq 1 \end{cases}$$

For an arithmetic sequence we have the explicit formula $a_n = a + nd$.

Definition 1.3: Geometric Sequence

A geometric sequence is defined as follows:

$$a_n = \begin{cases} a, & n = 0 \\ a_{n-1} \cdot q, & n \geq 1 \end{cases}$$

In this case, we aswell have an explicit formula: $a_n = a \cdot q^n$.

However looking at the fibonacci sequence, we see that it is different from these two. So we will do a mathematical trick to solve it. This trick is generalizing.

Definition 1.4:

Let $a_0, a_1 \in \mathbb{R}$ we define $a_n := a_{n-1} + a_{n-2}$ for $n \geq 2$.

Denote the sequence we get $\mathcal{F}_{a_0, a_1}$.

If $S = (s_0, s_1, \dots)$ is any sequence of numbers, we say that S is Fibonacci if there exists $a_0, a_1 \in \mathbb{R}$ such that $S = \mathcal{F}_{a_0, a_1}$. Denote by Fib the set of all Fibonacci sequences.

Fib has some algebraic structures.

Claim 1.5:

If $\mathcal{F}' = (a_0, a_1, \dots)$ and $\mathcal{F}'' = (b_0, b_1, \dots)$ are Fibonacci then

$$\mathcal{F}' + \mathcal{F}'' := (a_0 + b_0, a_1 + b_1, \dots, a_n + b_n, \dots)$$

is also Fibonacci.

Proof. Lets write $c_0 := a_0 + b_0, c_1 := a_1 + b_1$ and $c_n := a_n + b_n$. Then $\mathcal{F}' + \mathcal{F}'' = (c_0, c_1, \dots)$. To show this is a fibonacci sequence we need to show that $c_n = c_{n-1} + c_{n-2}$ for $n \geq 2$.

Indeed,

$$\begin{aligned} c_{n-1} + c_{n-2} &= (a_{n-1} + b_{n-1}) + (a_{n-2} + b_{n-2}) \\ &= (a_{n-1} + a_{n-2}) + (b_{n-1} + b_{n-2}) \\ &= a_n + b_n \\ &= c_n. \end{aligned}$$

□

This is somewhat special, as not every set of sequences is closed under addition. In fact, we showed that

$$\mathcal{F}_{a_0, a_1} + \mathcal{F}_{b_0, b_1} = \mathcal{F}_{a_0 + b_0, a_1 + b_1}.$$

But there is more structure.

Claim 1.6:

Let $\mathcal{A} = (a_0, a_1, \dots)$ be a Fibonacci sequence and let $\alpha \in \mathbb{R}$. Define $\alpha\mathcal{A} := (\alpha a_0, \alpha a_1, \dots)$. Then $\alpha\mathcal{A}$ is also Fibonacci. In fact, $\alpha\mathcal{F}_{a_0, a_1} = \mathcal{F}_{\alpha a_0, \alpha a_1}$.

Proof. We need to check that $\alpha a_n = \alpha a_{n-1} + \alpha a_{n-2}$ for $n \geq 2$. But this is "obviously" true, because we know that $a_n = a_{n-1} + a_{n-2}$. □

Note: The sequence $(0, 0, \dots)$ is also Fibonacci, it is actually the sequence $\mathcal{F}_{0,0}$.

In fact, **LINEAR COMBINATIONS** of Fibonacci are also Fibonacci.

$$\alpha\mathcal{F}_{a_0, a_1} + \beta\mathcal{F}_{b_0, b_1} = \mathcal{F}_{\alpha a_0 + \beta b_0, \alpha a_1 + \beta b_1},$$

for $\alpha, \beta \in \mathbb{R}$.

Corollary 1.7:

In order to find a formula for $\mathcal{F}_{a_0, a_1}$ it is enough to find a formula for $\mathcal{F}_{0,1}$ and $\mathcal{F}_{1,0}$.

Proof. This is because $\mathcal{F}_{a_0, a_1} = a_0\mathcal{F}_{1,0} + a_1\mathcal{F}_{0,1}$.

□

Out of curiosity, could it be that for some, very special $a_0, a_1 \neq (0,0)$, the sequence $\mathcal{F}_{a_0, a_1}$ is arithmetic or geometric?

Since arithmetic sequences are characterized by a constant difference $a_n = a_{n-1} + d$, we would need $a_{n-1} + a_{n-2} = a_{n-1} + d$ for $n \geq 2$. This means that the sequence would

need to be constant, i.e. $a_n = d$ for all $n \geq 1$. This cannot happen for any a_0, a_1 except $(0, 0)$.

Perhaps it could be geometric? Let's try to see if $(1, q, \dots)$ is Fibonacci for some $q \in \mathbb{R}$.

For such a sequence to be Fibonacci we need

$$q^n = q^{n-1} + q^{n-2} \text{ for } n \geq 2.$$

As $q \neq 0$ we get $q^2 = q + 1$. This is a quadratic equation with solutions

$$q = \frac{1 \pm \sqrt{5}}{2}.$$

Denote $\varphi := \frac{1+\sqrt{5}}{2}$ and $\psi := \frac{1-\sqrt{5}}{2}$.

We have found

$$\mathcal{F}_{1,\varphi} = (1, \varphi, \varphi^2, \varphi^3, \dots)$$

$$\mathcal{F}_{1,\psi} = (1, \psi, \psi^2, \psi^3, \dots)$$

Perhaps a linear combination of $\mathcal{F}_{1,\varphi}$ and $\mathcal{F}_{1,\psi}$ will give us $\mathcal{F}_{0,1}$?

We need to find $\alpha, \beta \in \mathbb{R}$ such that

$$\begin{cases} \alpha \cdot 1 + \beta \cdot 1 = 0 \\ \alpha\varphi + \beta\psi = 1 \end{cases}.$$

Solving this system of equations gives

$$\alpha = \frac{1}{\varphi - \psi} = \frac{1}{\sqrt{5}}, \quad \beta = \frac{1}{\psi - \varphi} = -\frac{1}{\sqrt{5}}.$$

So we found

$$\begin{aligned} \mathcal{F}_{0,1} &= \frac{1}{\sqrt{5}}\mathcal{F}_{1,\varphi} - \frac{1}{\sqrt{5}}\mathcal{F}_{1,\psi} \\ &= \frac{1}{\sqrt{5}} \left(\left(\frac{1+\sqrt{5}}{2} \right)^n - \left(\frac{1-\sqrt{5}}{2} \right)^n \right). \end{aligned}$$

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1.2 Logic and Math language

What is a mathematical/logical statement? A statement that can be either true or false, but not both.

Example 1.8:

A: "For every real number x , we have $x^2 \geq 0$ ".

B: "Every prime number $p \geq 3$ must be odd".

C: "Every odd number $p \geq 3$ must be prime".

A is True, B is True, C is False.

Negation of a statement: If A is a statement, then the statement "A is NOT true" is called the negation of A , denoted by $\neg A$ ("not A").¹

Such a statement can be represented in a truth table:

A	$\neg A$
T	F
F	T

¹Sometimes we write $\bar{A}$ instead of $\neg A$.

Definition 1.9: And operation

Let A, B be statements. Then the statement $A \wedge B$ is the statement "A is true AND B is true". Sometimes we write $A \& B$ instead.

The corresponding truth table is:

A	B	$A \wedge B$
T	T	T
T	F	F
F	T	F
F	F	F

Example 1.10:

For real numbers x , $(x^2=1) \wedge (x \geq 0)$ is true. This is equivalent to $x=1$.

Similarly we can define the "or" operation.

Definition 1.11: Or operation

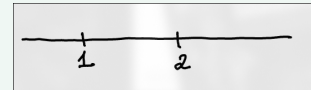
Let A, B be statements. Then the statement $A \vee B$ is the statement "A is true OR B is true", i.e. "at least one of the statements A or B are true". Sometimes we write $A|B$ instead.

Thus the corresponding truth table is:

A	B	$A \vee B$
T	T	T
T	F	T
F	T	T
F	F	F

Example 1.12:

$(x > 1) \vee (x < 2)$. We can draw a number line to visualize this situation:



We see that this statement is true for all $x \in \mathbb{R}$.

Example 1.13:

Consider the statement $(x < 1) \vee (x > 2)$. Again looking at the number line, we see that this statement is equivalent to $\neg(1 \leq x \leq 2)$.

Definition 1.14: Logical Implications

Let A, B be statements. We can form a new statement $A \Rightarrow B$ (A implies B). The statement " $A \Rightarrow B$ " is "If A is true, then B is also true".

Depicted in a truth table:

A	B	$A \Rightarrow B$
T	T	T
T	F	F
F	T	T
F	F	T

As an exercise: Convince yourself that the statement $A \Rightarrow B$ is equivalent to $(\neg A \vee B)$.

Example 1.15:

("0 = 1") $\Rightarrow$ ("Earth is Flat") is a true statement. Notice that in logical terms, implies does not mean because A is true, B is true, no matter if there is no causality.

Similarly there is logical equivalence:

Definition 1.16: Logical Equivalence

$A \Leftrightarrow B$ is the statement $(A \Rightarrow B) \wedge (B \Rightarrow A)$. In different words: A holds if and only if (iff) B holds.

We say that A and B are **EQUIVALENT STATEMENTS** if $A \Leftrightarrow B$ is true. This happens if A and B have the same truth table.

Example 1.17:

$$x^2 > 0 \Leftrightarrow x \neq 0,$$

$$0 = 1 \Leftrightarrow \text{Earth is flat.}$$

Let A, B be statements. Then there is a statement $A \Rightarrow B$ and $\neg B \Rightarrow \neg A$. These two statements are actually equivalent. This can be shown by truth tables.

As an exercise, state in words, why

$$\neg(\neg A) = A$$

and

$$\neg(A \wedge B) = (\neg A) \vee (\neg B)$$

Definition 1.18: Predicate

A **PREDICATE** is a statement involving some variables taken from a set.

$$P(x), P(x, y).$$

For example, $P(n) = "n = n^2"$.

"for all integers $n : n = n^2$ ". This is definitely a false statement. Meanwhile, "There exists an integer $n : n = n^2$ " is a true statement.

We denote $\forall$ for "for all/all" and $\exists$ for "there exists". The previous statements can be written as:

$$\forall n \in \mathbb{Z}, P(n) \quad , \quad \exists n \in \mathbb{Z}, P(n).$$

The symbols $\forall$ and $\exists$ are called **QUANTIFIERS**.

$\forall n \in \mathbb{Z} : n = n^2 \Rightarrow (n = 0) \vee (n = 1)$ is a true statement.

$\forall y \in \mathbb{R} : (y \geq 0) \Rightarrow (\exists x \in \mathbb{R} : x^2 = y)$ is true.

In practice, the second statement would be written as:

$$\forall 0 \leq y \in \mathbb{R}, \exists x \in \mathbb{R} \text{ s.t. } x^2 = y.$$

Be careful about the order of quantifiers:

$$\forall x \in X, \exists y \in Y : A(x, y),$$

is not the same as

$$\exists y \in Y, \forall x \in X : A(x, y).$$

Example 1.19:

Let X be the set of students at ETH and Y the set of all courses given in HS25. Let $A(x, y)$ be the statement "student x attends course y ". Then the first statement means "For every student, there is at least one course that the student attends". Which is true, whilst the second statement means "There is a course that every student attends", which is false.

In practice, instead of writing $\exists x \in X, \exists y \in X$, one rather writes $\exists x, y \in X$, similarly for $\forall$.

When negating quantifiers, we have the following rules:

$$\neg(\forall x \in X : A(x)) \Leftrightarrow \exists x \in X : \neg A(x).$$

$$\neg(\exists x \in X : A(x)) \Leftrightarrow \forall x \in X : \neg A(x).$$

There is also a quantifier "there exists a unique" denoted by $\exists!$.

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$$\exists! x \in X : A(x).$$

Example 1.20:

" $\exists! x \in \mathbb{R} : x^2 = 4$ ". This is a false statement as there exist two such x (namely 2 and -2).

" $\exists! x \in \mathbb{R} : x \geq 0 \wedge x^2 = 4$ ". This is a true statement as there exists exactly one such x (namely 2).

1.3 Set theory

The fundamental object in set theory is the set.

Definition 1.21: Set

A **SET** is a collection of different objects. These objects are called the **ELEMENTS** of the set.

For example, $M = \{1, 3, \text{unicorn}\} = \{3, 1, \text{unicorn}\}$ is a set with three elements. As we can see, the order of elements does not matter. Furthermore, repetition of elements does not matter, i.e. $\{1, 1, 1\} = \{1\}$.

If x is an element of a set M , we write $x \in M$. If it is not, we write $x \notin M$.

Another way to write sets is by specifying a property that the elements satisfy.

$$M = \{x : A(x)\} = \{x \mid A(x)\},$$

represents the set of all x such that $A(x)$ holds.

Example 1.22:

$$\{x \mid x \in \mathbb{Z} \wedge x^2 \leq 9\} = \{-3, -2, -1, 0, 1, 2, 3\}.$$

Another important set is the empty set $\{\}$, which contains no elements. It is denoted by $\emptyset$.

A set is allowed to contain other sets as elements one of its elements. For example $M = \{1, \{2, 3\}\}$ is a set with two elements, 1 and $\{2, 3\}$. Similarly, $\{\emptyset\}$ is a set with one element, the empty set.

Definition 1.23:

Let P, Q be sets.

1. We say that P is a **SUBSET** of Q if $x \in P \Rightarrow x \in Q$. We write $P \subseteq Q$ (Also: $P \subset Q$).
2. $P \subsetneq Q$ means $P \subseteq Q$ but $P \neq Q$.
3. $P \not\subseteq Q$ means $\neg(P \subseteq Q)$.

Careful: Every set is a subset of itself. Furthermore, the empty set $\emptyset \subseteq M$ for every set M .

We define the following operations for sets.

$$\begin{aligned} P \cap Q &:= \{x : x \in P \wedge x \in Q\} && \text{(intersection)} \\ P \cup Q &:= \{x : x \in P \vee x \in Q\} && \text{(union)} \\ P \setminus Q &:= \{x : x \in P \wedge x \notin Q\} && \text{(complement of } Q \text{ in } P) \\ P \Delta Q &:= (P \cup Q) \setminus (P \cap Q) && \text{(symmetric difference)} \end{aligned}$$

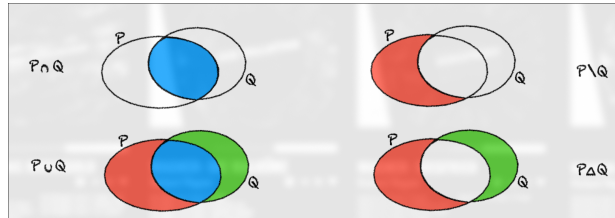


Figure 1: Set operations

Sometimes our sets will be subsets of some ambient/underlying set X .

In this case we define the **COMPLEMENT** of P in X as

$$P^c := X \setminus P = \{x \in X : x \notin P\}.$$

Definition 1.24:

Let $\mathcal{A}$ be a family of sets, $\mathcal{A} \neq \emptyset$. We can look at the union of the members of $\mathcal{A}$:

$$\bigcup_{A \in \mathcal{A}} A := \{x \mid \exists A \in \mathcal{A} \text{ s.t. } x \in A\}.$$

Similarly, we can look at the intersection of the members of $\mathcal{A}$:

$$\bigcap_{A \in \mathcal{A}} A := \{x \mid \forall A \in \mathcal{A} : x \in A\}.$$

When working with big unions and intersections, it is often useful to translate them into set statements involving quantifiers, as this greatly improves intuition.

Example 1.25:

Let $\mathcal{A} = \{2, 3, 4, \dots\}$.

Let $A_n = \{x \mid x \in \mathbb{Z} \geq 1, n^2 \mid x\}$.

Then

$$\bigcup_{n \in \mathcal{A}} A_n = \{a \in \mathbb{Z} \mid \exists k \geq 2, \exists r \in \mathbb{Z} \text{ s.t. } r \geq 1 \wedge a = k^2 r\}$$

and

$$\bigcap_{n \in \mathcal{A}} A_n = \{\}.$$

Definition 1.26: Cartesian Product

Let X, Y be sets. The **CARTESIAN PRODUCT** of X and Y is the set of ordered pairs (x, y) .

$$X \times Y := \{(x, y) \mid x \in X, y \in Y\}.$$

Definition 1.27: n-Fold Cartesian Product

Let X be a set, $n \geq 1$. The **n-FOLD CARTESIAN PRODUCT** of X is defined as

$$X^n := \underbrace{X \times X \times \dots \times X}_{n \text{ times}}.$$

An element of $X^n = (x_1, x_2, \dots, x_n)$ is called an **N-TUPLE**.

The classic example is $\mathbb{R}^2 = \mathbb{R} \times \mathbb{R}$ – the Cartesian plane.

Definition 1.28: Power Set

Let X be a set. The **POWER SET** of X is the set of all subsets of X .

$$\mathcal{P}(X) := \{A \mid A \subseteq X\}.$$

Sometimes the notation 2^X is also used.

Example 1.29:

Let $X = \{1, 2, 3\}$. Then

$$\begin{aligned} \mathcal{P}(X) = \{ &\emptyset, \{1\}, \{2\}, \{3\}, \\ &\{1, 2\}, \{1, 3\}, \{2, 3\}, \\ &\{1, 2, 3\} \} \end{aligned}$$

Notice that if X has n elements, then $\mathcal{P}(X)$ has 2^n elements.

The **CARDINALITY** of a set X is the number of elements in X . For example, if $X = \{1, 2, \dots, n\}$, then $|X| = n$. This operation is only defined for finite sets at the moment.

1.4 Functions/Maps

Definition 1.30: Maps

Let X, Y be sets. A **MAP** $f : X \rightarrow Y$ is an assignment to *every* element $x \in X$ a uniquely defined element $f(x) \in Y$.

Roughly speaking, for an input $x \in X$, the map f produces an output $f(x) \in Y$.

Given a map $f : X \rightarrow Y$, The set of inputs of a map is called the **DOMAIN OF DEFINITION** of f , whilst Y is called the **TARGET** of f .

Given two functions $f : X \rightarrow Y$ and $g : X' \rightarrow Y'$, we say that f and g are equal if $X = X'$, $Y = Y'$ and $\forall x \in X : f(x) = g(x)$.

Example 1.31:

$$f(x) := x^2.$$

This is lacking some information. Much better would be

$$X := \{x \in \mathbb{R} \mid 0 \leq x\}, \quad f : X \rightarrow \mathbb{R}, \quad f(x) := x^2.$$

Here is another function:

$$Z := \{x \in \mathbb{R} \mid -12 \leq x\}, \quad g : Z \rightarrow \mathbb{R}, \quad g(x) := x^2.$$

According to our definition, $f \neq g$ since $X \neq Z$.

Functions can be described by a formula, but do not necessarily have to be.

About notation: $f : X \rightarrow Y$ is the most standard notation. Sometimes $X \ni x \rightarrow f(x) \in Y$ is used.

Definition 1.32:

Let $f : X \rightarrow Y$ be a map. The **IMAGE** of f is the set

$$f(X) := \{y \in Y \mid \exists x \in X : f(x) = y\}.$$

Sometimes it's also written as $\text{image}(f)$. It holds that $f(X) \subseteq Y$.

Definition 1.33: Restriction

Let $f : X \rightarrow Y$, Let $A \subseteq X$. We can define a new map

$$f|_A : A \rightarrow Y, \quad f|_A(a) = f(a) \quad \forall a \in A$$

called the **RESTRICTION** of f to A .

A restriction restricts the domain of definition of a function.

Example 1.34: Characteristic Function

Let X be a set, $A \subseteq X$. The **CHARACTERISTIC FUNCTION** of A is defined as

$$\mathbb{1}_A : X \rightarrow \{0, 1\}, \quad \mathbb{1}_A(x) := \begin{cases} 1 & x \in A \\ 0 & x \notin A \end{cases}.$$

Example 1.35: Projection Function

Let X, Y be sets. The **PROJECTION FUNCTION** on X is defined as

$$\pi_X : X \times Y \rightarrow X, \quad \pi_X(x, y) := x.$$

Similarly, we can define $\pi_Y : Y \times X \rightarrow Y$.

Functions can have different properties.

Definition 1.36:

Let $f : X \rightarrow Y$ be a map.

- f is called **INJECTIVE** if

$$\forall x_1, x_2 \in X : f(x_1) = f(x_2) \Rightarrow x_1 = x_2.$$

Equivalently, if $x_1 \neq x_2$, then $f(x_1) \neq f(x_2)$.

- f is called **SURJECTIVE** if

$$\forall y \in Y, \exists x \in X : f(x) = y.$$

Equivalently, $f(X) = Y$.

- f is called **BIJECTIVE** if it is both injective and surjective. Equivalently,

$$\forall y \in Y, \exists! x \in X : f(x) = y.$$

Definition 1.37:

Let $f : X \rightarrow Y$ be a bijection. The **INVERSE** map $f^{-1} : Y \rightarrow X$ is the map that assigns to every $y \in Y$ the unique $x \in X$ such that $f(x) = y$.

An example would be the map $f : \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}_{\geq 0}, f(x) = x^2$. Its inverse is $f^{-1} : \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}_{\geq 0}, f^{-1}(y) = \sqrt{y}$.

Definition 1.38: Composition

Let $f : X \rightarrow Y, g : Y \rightarrow Z$ be maps. We define $g \circ f : X \rightarrow Z$ as the **COMPOSITION** of f and g :

$$g \circ f(x) := g(f(x)).$$

Sometimes this is written as

$$X \xrightarrow{f} Y \xrightarrow{g} Z.$$

Example 1.39:

Let $f : X \rightarrow Y$ be a bijection and $f^{-1} : Y \rightarrow X$ its inverse map. We have $f \circ f^{-1} = \text{id}_Y$ and $f^{-1} \circ f = \text{id}_X$, where $\text{id}_X : X \rightarrow X, \text{id}_X(x) = x$ is the identity map on X .

The order of composition is important. In general, $f \circ g$ and $g \circ f$ are different maps. Even worse, one of them might not even be defined.

Lemma 1.40:

Let $f : X \rightarrow Y, g : Y \rightarrow Z$ be maps.

1. If f, g are injective, then $g \circ f$ is injective.
2. If f, g are surjective, then $g \circ f$ is surjective.
3. If f, g are bijective, then $g \circ f$ is bijective and

$$(g \circ f)^{-1} = f^{-1} \circ g^{-1}.$$

Proof.

1. Let $x_1, x_2 \in X$ such that $(g \circ f)(x_1) = (g \circ f)(x_2)$. Then $g(f(x_1)) = g(f(x_2))$. Since g is injective, $f(x_1) = f(x_2)$. Since f is injective, $x_1 = x_2$. Thus $g \circ f$ is injective.
2. Let $z \in Z$. Since g is surjective, there exists $y \in Y$ such that $g(y) = z$. Since f is surjective, there exists $x \in X$ such that $f(x) = y$. Thus $(g \circ f)(x) = g(f(x)) = g(y) = z$. Hence $g \circ f$ is surjective.
3. Follows from 1. and 2.

□

This lemma is useful when showing a certain map is injective/surjective/bijective, as it allows us to work with simpler maps.

Definition 1.41: Image

Let $f : X \rightarrow Y$ be a map, $A \subseteq X$. Then

$$f(A) := \{y \in Y \mid \exists x \in A : f(x) = y\}$$

is called the **IMAGE** of A under f .

Definition 1.42: Inverse Image

Let $B \subseteq Y$. Define the **INVERSE IMAGE** of B under f as

$$f^{-1}(B) := \{x \in X \mid f(x) \in B\}.$$

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1.4.1 Graphs

Consider the following.

Definition 1.43:

Let $f : X \rightarrow Y$ be a map. The **graph** of f is the subset $\text{graph}(f) \subseteq X \times Y$ defined by

$$\text{graph}(f) := \{(x, f(x)) \mid x \in X\}.$$

Example 1.44:

If $X \subseteq \mathbb{R}$ and $Y = \mathbb{R}$, then the graph can be visualized in the Cartesian plane $\mathbb{R}^2$.

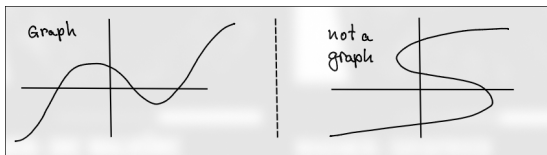


Figure 2: Graphs of some functions $f : \mathbb{R} \rightarrow \mathbb{R}$.

Sometimes injectivity and surjectivity can be read off from the graph.

2 Linear Systems of Equations

2.1 Fields

A **FIELD** (Körper) is a set K with two operations $+$ and $\cdot$. Together they form a commutative group $(K, +)$ with identity 0 and inverses $-a$ for $a \in K$. The operation $\cdot$ is associative and commutative with identity $1 \neq 0$ and inverses a^{-1} for $a \in K \setminus \{0\}$. The operations are linked by the distributive law

$$a \cdot (b + c) = a \cdot b + a \cdot c.$$

Furthermore, we ask that $1 \neq 0$.

Example 2.1: Examples of fields

$$\mathbb{Q} = \left\{a \mid a = \frac{m}{n}, m, n \in \mathbb{Z}, n \neq 0\right\}$$

with the usual addition and multiplication is a field.

$\mathbb{R}$ = the real numbers.

$$\mathcal{F}_2 = \{0, 1\},$$

with the following addition and multiplication tables:

$$\begin{array}{c|cc} + & 0 & 1 \\ \hline 0 & 0 & 1 \\ 1 & 1 & 0 \end{array} \quad \begin{array}{c|cc} \cdot & 0 & 1 \\ \hline 0 & 0 & 0 \\ 1 & 0 & 1 \end{array}.$$

Similarly, we can define $\mathcal{F}_3 = \{0, 1, 2\}$ and $\mathcal{F}_5 = \{0, 1, 2, 3, 4\}$, where the operations are defined modulo 3 and 5, respectively.

2.2 Systems of Linear Equations

Consider the following

$$\begin{cases} x + 3y + 4z = -1 \\ 2x - y + z = 5 \end{cases}.$$

To solve this, there exists a systematic method which is called **ELIMINATION**.

$$\begin{aligned} \begin{cases} x + 3y + 4z = -1 \\ 2x - y + z = 5 \end{cases} & \xrightarrow[-\rightarrow E_2]{-2E_1 + E_2} \begin{cases} x + 3y + 4z = -1 \\ -7y - 7z = 7 \end{cases} \\ & \xrightarrow[-\rightarrow E_2]{-\frac{1}{7}E_2} \begin{cases} x + 3y + 4z = -1 \\ y + z = -1 \end{cases} \\ & \xrightarrow[-\rightarrow E_1]{-3E_2 + E_1} \begin{cases} x + z = 2 \\ y + z = -1 \end{cases} \end{aligned}$$

Staring at this for a while, we can see that z is a free variable. Thus, the set of solutions is given by:

choose z arbitrarily, $z = c \in \mathbb{R}$

then $y = -1 - c$ and $x = 2 - c$

$$(x, y, z) = (2 - c, -1 - c, c), c \in \mathbb{R} \subseteq \mathbb{R}^3.$$

A general system of many equations and matrix notation:

Fix a field $\mathcal{F}$. A system of m equations in n unknowns over $\mathcal{F}$ with unknowns $x_1, \dots, x_n$ is of the form

$$\begin{cases} a_{11}x_1 + a_{12}x_2 + \dots + a_{1n}x_n &= b_1 \\ a_{21}x_1 + a_{22}x_2 + \dots + a_{2n}x_n &= b_2 \\ \vdots &\vdots \\ a_{m1}x_1 + a_{m2}x_2 + \dots + a_{mn}x_n &= b_m \end{cases}.$$

We call $a_{ij} \in \mathcal{F}$ the coefficients. Furthermore $b_i \in \mathcal{F}$. Such a system is called **LINEAR** because the unknowns appear only to the first power and are not multiplied together.

We notice the objects can be written as matrices and vectors.

$$A = \begin{pmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{pmatrix}.$$

This object is called an $m \times n$ matrix with entries in $\mathcal{F}$.

Sometimes we write $A = (a_{ij})_{1 \leq i \leq m, 1 \leq j \leq n}$.

Further we organize

$$x = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix}, \quad b = \begin{pmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{pmatrix}.$$

These are $n \times 1$ **VECTORS**.

The system can now be written as

$$(S) : A \cdot x = b.$$

And the solution set as

$$L(S) = \{(x_1, x_2, \dots) \mid x_i \in \mathcal{F} \forall 1 \leq i \leq n \wedge A \cdot x = b\}.$$

We define the matrix multiplication as follows:

$$\begin{pmatrix} a_{11} & \dots & a_{1n} \\ \vdots & \ddots & \vdots \\ a_{m1} & \dots & a_{mn} \end{pmatrix} \cdot \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} = \begin{pmatrix} a_{11}x_1 + \dots + a_{1n}x_n \\ \vdots \\ a_{m1}x_1 + \dots + a_{mn}x_n \end{pmatrix}.$$

If we now have $A \cdot x = b$, this is equivalent to the system of equations. We call this the **MATRIX NOTATION**.

Example 2.2:

The system from the beginning can be written as

$$\begin{pmatrix} 1 & 3 & 4 \\ 2 & -1 & 1 \end{pmatrix} \cdot \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} -1 \\ 5 \end{pmatrix}.$$

We can also write the **EXTENDED MATRIX** as

$$(A|b) = \left(\begin{array}{ccc|c} 1 & 3 & 4 & -1 \\ 2 & -1 & 1 & 5 \end{array} \right).$$

2.3 Elementary Row Operations

We define the following operations on matrices.

1. Choose $c \neq 0 \in \mathcal{F}$; multiply eq./row i by c .
($c \cdot R_i \rightarrow R_i$)
2. Choose $c \in \mathcal{F}$, choose $1 \leq i, j \leq m, i \neq j$; replace eq./row j by eq./row $j + c \cdot$ eq./row i .
($c \cdot R_i + R_j \rightarrow R_j$)
3. Interchange eq./row i and eq./row j .
($R_i \leftrightarrow R_j$)

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Definition 2.3: Row-Equivalence

Let C and C' be $r \times s$ matrices with entries in $\mathcal{F}$. We say that C' is **ROW-EQUIVALENT** to C if C' can be obtained from C by a finite sequence of elementary row operations.

If we denote $C = C_0 \rightarrow C_1 \rightarrow \dots \rightarrow C_k = C'$, then the following holds

Theorem 2.4:

Let $A \cdot x = b$ and $A' \cdot x = b'$ be two systems of m linear equations in n unknowns. Let's call first system (S) and second system (S'). Suppose the extended matrix $(A'|b')$ is row-equivalent to $(A|b)$. Then

$$L(S') = L(S).$$

Proof. We begin with the following claim:

Claim 1: If S' is obtained from S by one elementary row operation, then $L(S) \subseteq L(S')$.

Let $(x_1, \dots, x_n) \in L(S)$. We need to show that $(x_1, \dots, x_n) \in L(S')$.

Operation 1: $c(\neq 0) \cdot R_i \rightarrow R_i$. In this case all the equations of S' coincide with the equations of S except equation i .

$$a_{i1}x_1 + a_{i2}x_2 + \dots + a_{in}x_n = b_i$$

$$\xrightarrow{c \cdot R_i \rightarrow R_i} ca_{i1}x_1 + ca_{i2}x_2 + \dots + ca_{in}x_n = cb_i.$$

If $(x_1, \dots, x_n)$ satisfies the first equation, it also satisfies the second equation, since we can multiply both sides of the first equation by c .

Operation 2: $R_j + c \cdot R_i \rightarrow R_j$. Again, all equations of S and S' coincide except equation j . We have

$$a_{j1}x_1 + \dots + a_{jn}x_n = b_j$$

$$\begin{aligned} &\xrightarrow{R_j + c \cdot R_i \rightarrow R_j} (a_{j1} + ca_{i1})x_1 + \dots + (a_{jn} + ca_{in})x_n = b_j + cb_i \\ &\Rightarrow (a_{j1}x_1 + \dots + a_{jn}x_n) + c(a_{i1}x_1 + \dots + a_{in}x_n) \\ &= b_j + cb_i. \end{aligned}$$

So if $(x_1, \dots, x_n)$ satisfies the first equation, it also satisfies the second equation, since we can add c times the i -th equation to the j -th equation.

Operation 3: $R_i \leftrightarrow R_j$. In this case, clearly we have that

$$(x_1, \dots, x_n) \in L(S) \Rightarrow (x_1, \dots, x_n) \in L(S').$$

This concludes the proof of Claim 1.

Claim 2: If S' is obtained from S by one elementary row operation, then S can be obtained from S' by one elementary row operation.

(1): If $S \xrightarrow[c \neq 0]{c \cdot R_i \rightarrow R_i} S'$, then $S' \xrightarrow{\frac{1}{c} \cdot R_i \rightarrow R_i} S$.

(2): If $S \xrightarrow[i \neq j]{R_j + c \cdot R_i \rightarrow R_j} S'$, then $S' \xrightarrow{R_j - c \cdot R_i \rightarrow R_j} S$.

(3): If $S \xrightarrow{R_i \leftrightarrow R_j} S'$, then $S' \xrightarrow{R_i \leftrightarrow R_j} S$.

This concludes the proof of Claim 2.

Claim 3: If S' is obtained from S by one elementary row operation, then $L(S) = L(S')$.

By Claim 1, we have $L(S) \subseteq L(S')$. By Claim 2, S is obtained by S' by one elementary row operation. Hence by Claim 1 again (with the roles of S and S' interchanged), we have $L(S') \subseteq L(S)$.

It follows that $L(S) = L(S')$.

Proof of Theorem: We are now in position to prove the theorem. By assumption, there is a finite sequence of elementary row operations

$$S = S_0 \rightarrow S_1 \rightarrow \dots \rightarrow S_k = S'.$$

By Claim 3, we have $L(S_0) = L(S_1) = \dots = L(S_k)$, i.e.

$$L(S) = L(S').$$

□

Definition 2.5: Row-Reduced Matrix

A $m \times n$ matrix A is called **ROW-REDUCED** if the following two conditions hold:

1. The first non-zero entry in each non-zero row of A is 1. This entry is called the **LEADING ENTRY/PIVOT** of that row.
2. Each column of A which contains the leading non-zero entry of some row has all its other entries equal to 0.

Example 2.6:

$$A = \begin{pmatrix} 0 & 0 & 0 & 1 & 2 \\ 0 & 1 & -3 & 0 & \frac{1}{2} \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}.$$

This matrix is row-reduced.

$$B = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 1 & 0 \end{pmatrix}.$$

This matrix is not row-reduced, because the second condition is violated in column 3.

$$C = \begin{pmatrix} 0 & 2 & 1 \\ 1 & 0 & -3 \\ 0 & 0 & 0 \end{pmatrix}.$$

This matrix is not row-reduced, because the first condition is violated in row 1.

It might not be clear yet, why row-reduced matrices are important, however, we will see this soon. However, we will first prove the following theorem.

Theorem 2.7:

Every $m \times n$ matrix A is row-equivalent to a row-reduced $m \times n$ matrix.

Proof. Let $A = (a_{ij})_{1 \leq i \leq m, 1 \leq j \leq n}$, with $a_{ij} \in \mathcal{F}$.

If all entries in the 1st row of A are 0, the condition (1) is satisfied for row 1.

If row 1 does have a non-zero entry, let k be the smallest index j for which $a_{1j} \neq 0$. Multiply row 1 by a_{1k}^{-1} to make the leading entry 1. Now, condition (1) is satisfied for row 1.

For each row $2 \leq i \leq m$ add $(-a_{ik})$ times row 1 to row i . Formally

$$R_i + (-a_{ik}) \cdot R_1 \rightarrow R_i.$$

The result of these operations is a matrix A' which looks like

$$\begin{pmatrix} 0 & \dots & 0 & 1 & \dots \\ \dots & \dots & \dots & 0 & \dots \\ \vdots & \ddots & \vdots & \vdots & \vdots \\ a_{i1} & \dots & a_{ik-1} & 0 & \dots \\ \vdots & \vdots & \vdots & \vdots & \vdots \end{pmatrix}.$$

Notice that in row i of resulting matrix, the elements standing to the right of entry k (which is now 0) are unchanged. This is because to the left of the pivot in row 1, there are only 0s.

Summary: Condition (2) is now satisfied for the column of the pivot of row 1.

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We now turn to row number 2. If all elements in row 2 are 0, we just leave it as it is. In case not all elements are 0, we find the pivot in that row, say at entry $2, k'$.

Note that $k' \neq k$, because in column k of row 2, we have 0. Divide row 2 by the element lying at $(2, k')$ to make the pivot equal to 1.

Now we take row two and add suitable multiples of it to every other zero. Notice that for the critical row 1, this operation does not change the pivot, because $(2, k) = 0$. To prove that the pivot stays in row 1, we distinguish two cases:

If $k' < k$, the entry in the first row is already 0, so the multiple is $c = 0$.

If $k' > k$, by definition of k' , all entries in row 2 to the left of k' and thus also k are 0. Thus they won't change.

We can continue this process for rows $3, 4, \dots, m$. In the end, we arrive at a row-reduced matrix. □

Tip 2.8:

Only divide by pivot elements at the end of the algorithm, to avoid fractions.

Definition 2.9:

An $m \times n$ matrix is called **ROW-REDUCED ECHELON** if the following holds:

- (a) It is row-reduced.
- (b) Every row which has only 0 entries appears below every row which has a non-zero entry.
- (c) If rows $1, \dots, r$ are the non-zero rows, and if the pivot of row i occurs in column k_i , then

$$k_1 < k_2 < \dots < k_r.$$

A row-reduced echelon matrix looks as follows:

$$\begin{pmatrix} 0 & \dots & 0 & 1 & * & 0 & * & 0 & * \\ 0 & \dots & 0 & 0 & 0 & 1 & * & 0 & * \\ 0 & \dots & 0 & 0 & 0 & 0 & 0 & 1 & * \\ 0 & \dots & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}.$$

Theorem 2.10:

Every $m \times n$ matrix is row equivalent to a row reduced echelon matrix.

Proof. Apply the previous theorem + permutation between the rows. □

This is useful, for solving $A \cdot x = b$. Take $(A|b)$ and apply a sequence of row operations to bring it to row-reduced echelon form $(A'|b')$. Then $L(A|b) = L(A'|b')$.

Example 2.11:

We solve the following system of equations:

$$\begin{aligned} -9x_2 + 3x_3 + 4x_4 &= 9 \\ x_1 + 4x_2 - x_4 &= 5 \\ 2x_1 + 6x_2 - x_3 + 5x_4 &= -5. \end{aligned}$$

We transform this into an extended matrix

$$(A|b) = \left(\begin{array}{cccc|c} 0 & -9 & 3 & 4 & 9 \\ 1 & 4 & 0 & -1 & 5 \\ 2 & 6 & -1 & 5 & -5 \end{array} \right).$$

The corresponding row reduced matrix is

$$(A'|b') = \left(\begin{array}{cccc|c} 0 & 1 & 0 & -\frac{5}{3} & \frac{12}{5} \\ 1 & 0 & 0 & \frac{17}{3} & -\frac{23}{5} \\ 0 & 0 & 1 & -\frac{11}{3} & \frac{51}{5} \end{array} \right).$$

Thus the row reduced echelon is

$$\left(\begin{array}{cccc|c} 1 & 0 & 0 & \frac{17}{3} & -\frac{23}{5} \\ 0 & 1 & 0 & -\frac{5}{3} & \frac{12}{5} \\ 0 & 0 & 1 & -\frac{11}{3} & \frac{51}{5} \end{array} \right).$$

We can see that we can take $x_4 := a \in \mathbb{R}$ as a free variable. Then we have

$$x_3 = \frac{11}{3}a + \frac{51}{5}, \quad x_2 = \frac{5}{3}a + \frac{12}{5}, \quad x_1 = -\frac{17}{3}a - \frac{23}{5}.$$

It may happen that the system has a row with only zeros. For example

$$0 = b_3 - b_2 + 2b_1.$$

Then if $b_3 - b_2 + 2b_1 \neq 0$, the system has no solution. Otherwise, the system is equivalent to the system without that equation.

3 Vector Spaces

Vector spaces are essentially the playground of linear algebra, the following definition is thus very central for the subject.

Definition 3.1: Vector Space

A **VECTOR SPACE** over a field $\mathcal{F}$ (scalars) is a set V (vectors) endowed with two operations:

$$\begin{aligned} + : V \times V &\rightarrow V, (v_1, v_2) \mapsto v_1 + v_2 \\ \cdot : \mathcal{F} \times V &\rightarrow V, (\lambda, v) \mapsto \lambda \cdot v. \end{aligned}$$

such that the following axioms are satisfied:

V1-4) $(V, +)$ is an abelian group.

$$\text{V5) } \forall a, b \in \mathcal{F}, \forall v \in V : a \cdot (b \cdot v) = (ab) \cdot v$$

$$\text{V6) } \forall v \in V, 1 \cdot v = v$$

$$\text{V7) } \forall a \in \mathcal{F}, v_1, v_2 \in V : a \cdot (v_1 + v_2) = a \cdot v_1 + a \cdot v_2$$

$$\text{V8) } \forall a_1, a_2 \in \mathcal{F}, v \in V : (a_1 + a_2) \cdot v = a_1 \cdot v + a_2 \cdot v$$

Tip 3.2:

Vector spaces are essentially **ORDERED LISTS** with component-wise operations.

These lists don't have to be finite.

Example 3.3: Coordinate Space

The coordinate space K^n for $n \in \mathbb{N}_0$ then

$$\underbrace{K^n}_{K \times K \cdots \times K} = \{(a_1, \dots, a_n) \mid a_i \in K, 1 \leq i \leq n\}.$$

We turn $V = K^n$ into a vector space. Let $v = (a_1, \dots, a_n), w = (b_1, \dots, b_n) \in V$. Then

$$v + w = (a_1 + b_1, a_2 + b_2, \dots, a_n + b_n) \in V.$$

Let $a \in K$, then

$$a \cdot v := (a \cdot a_1, a \cdot a_2, \dots, a \cdot a_n) \in V.$$

$$0 := (0, 0, \dots, 0) \in V.$$

As an exercise, show that K^n with these operations is a vector space.

Example 3.4: Matrix Space

Consider $M_{m \times n}$. The elements are

$$\begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{pmatrix} \quad a_{ij} \in K.$$

This is basically the same as $K^{m \cdot n}$.

Lemma 3.5:

Let V be a vector space over K . Then:

- (a) The zero vector is unique.
- (b) Let $v \in V$. The element $v' \in V$ with $v + v' = 0$ is unique. It is denoted by $-v$.
- (c) $\forall v \in V : 0 \cdot v = 0_V$
- (d) $\forall a \in K : a \cdot 0_V = 0_V$
- (e) $\forall v \in V : (-1) \cdot v = -v$
- (f) $\forall v \in V, -(-v) = v$
- (g) If $a \cdot v = 0$ for some $a \in K, v \in V$, then either $a = 0$ or $v = 0_V$.

Proof. (a): Since $(V, +)$ is an abelian group, there is a unique identity element, which we denote by 0_V .

(b): Again, since $(V, +)$ is an abelian group, every element has a unique inverse.

(c): Let $v \in V$. We have to show that $0 \cdot v = 0_V$. Indeed,

$$0 \cdot v = (0 + 0) \cdot v \stackrel{V8}{=} 0 \cdot v + 0 \cdot v.$$

Add $(-0 \cdot v)$ to both sides. This yields

$$0_V = 0 \cdot v + (0 \cdot v + (-0 \cdot v)) = 0 \cdot v + 0_V = 0 \cdot v.$$

(g): Let $a \in K, v \in V$ such that $a \cdot v = 0_V$. We need to prove that either $a = 0$ or $v = 0_V$. Indeed, assume that $a \neq 0$. By the axioms of a field, $\exists$ an element $a^{-1} \in K$ such that $a^{-1}a = 1 = aa^{-1}$.

Now:

$$v \stackrel{V6}{=} 1 \cdot v = (a^{-1}a) \cdot v \stackrel{V5}{=} a^{-1} \cdot (a \cdot v) = a^{-1} \cdot 0_V \stackrel{(d)}{=} 0_V.$$

□

Definition 3.6: Linear Subspace

Let V be a vector space over K . A subset $W \subseteq V$ is called a **LINEAR SUBSPACE** (Unterraum) of V if the following holds:

- LSS1) $W \neq \emptyset$,
- LSS2) $\forall w_1, w_2 \in W : w_1 + w_2 \in W$,
- LSS3) $\forall w \in W, \forall a \in K : a \cdot w \in W$.

Lemma 3.7:

Let V be a vector space over K and $W \subseteq V$ a subspace. Then W is a vector space on its own when endowed with the operations coming from V .

Proof. From LSS2 and LSS3, it follows that the operations $+$ and $\cdot$ from V indeed define such operations on W .

Now we have to check the axioms V1-V8. This can be done as an exercise.

For example, for V2, Pick any $w \in W$ (Possible by LSS1). We have

$$0_V \stackrel{(c)}{=} 0_K \cdot w \stackrel{\text{LSS3}}{\in} W.$$

Thus $0_V \in W$. Take now $0_W := 0_V$. Clearly, $\forall w \in W$ we have

$$0_W + w = 0_V + w \stackrel{V2}{=} w.$$

□

The above definition is often cumbersome to use. The following lemma gives a more handy criterion to check if a subset is a subspace.

Lemma 3.8: Criterion for Subspaces

Let V be a vector space over K and $W \subseteq V$ a subset. Then W is a subspace of V iff the following holds:

- (1) $0_V \in W$,
- (2) $\forall a_1, a_2 \in K, \forall w_1, w_2 \in W : a_1 \cdot w_1 + a_2 \cdot w_2 \in W$.

Proof. $\Rightarrow$: Assume that W is a subspace of V . This follows at once from the fact that W is in itself a vector space.

$\Leftarrow$: Assume that (1) and (2) hold. We have to show that W is a subspace. Clearly, LSS1 holds by (1). To show LSS2, pick $w_1, w_2 \in W$. Then by (2) (with $a_1 = a_2 = 1$), we have

$$w_1 + w_2 = 1 \cdot w_1 + 1 \cdot w_2 \in W.$$

Similarly for LSS3, pick $w \in W, a \in K$. Then by (2) (with $a_1 = a, a_2 = 0, w_1 = w, w_2 = 0_V$), we have

$$a \cdot w = a \cdot w + 0 \cdot 0_V \in W.$$

□

Example 3.9: Trivial Subspace

Let V be a vector space over K . Then

$$\{0_V\} \subseteq V,$$

is a subspace of V . It is called the **ZERO SUBSPACE** of V .

Example 3.10:

Fix $b \in K$. Consider the subset

$$\mathcal{U}_b := \{(x_1, x_2, x_3) \in K^3 \mid x_1 + x_2 + x_3 = b\} \subseteq K^3.$$

Claim: $\mathcal{U}_b$ is a subspace of K^3 iff $b = 0$.

Proof. $\Rightarrow$: Suppose $\mathcal{U}_b$ is a linear subspace of K^3 . If (x_1, x_2, x_3) and $(y_1, y_2, y_3) \in \mathcal{U}_b$, then

$$x_1 + x_2 + x_3 = b = y_1 + y_2 + y_3.$$

Since $\mathcal{U}_b$ is a linear subspace, we have

$$(x_1 + y_1, x_2 + y_2, x_3 + y_3) \in \mathcal{U}_b.$$

And thus it follows that

$$(x_1 + y_1) + (x_2 + y_2) + (x_3 + y_3) = b.$$

But from before, we get that this sum is equal to $2b$. Thus $2b = b$, which implies that $b = 0$.

The other direction is left as an exercise. □

Let A be an $m \times n$ matrix with entries in K . From now on we abbreviate this with

$$A \in \text{Mat}_{m \times n}(K) \text{ or } A \in M_{m \times n}(K).$$

Fix $b \in K^m$ such that $b = \begin{pmatrix} b_1 \\ \vdots \\ b_m \end{pmatrix}$ is viewed as a column vector. Consider the subset of K^n defined by

$$L := \{x \in K^n \mid A \cdot x = b\}.$$

Claim 3.11:

$L \subseteq K^n$ is a linear subspace iff $b = 0_V$.

For the proof we need preparations.

Lemma 3.12:

$\forall a \in K, u, v \in K^n$ we have

- (1) $A \cdot (u + v) = A \cdot u + A \cdot v$,
- (2) $A \cdot (a \cdot u) = a \cdot (A \cdot u)$.

Proof. Write $A = (a_{ij})_{1 \leq i \leq m, 1 \leq j \leq n}$,

$$A = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{pmatrix}, u = \begin{pmatrix} u_1 \\ u_2 \\ \vdots \\ u_n \end{pmatrix}, v = \begin{pmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{pmatrix}.$$

$$A \cdot u + A \cdot v =$$

$$\begin{pmatrix} \vdots \\ a_{i1}u_1 + a_{i2}u_2 + \cdots + a_{in}u_n \\ \vdots \end{pmatrix} + \begin{pmatrix} \vdots \\ a_{i1}v_1 + a_{i2}v_2 + \cdots + a_{in}v_n \\ \vdots \end{pmatrix} \\ = \begin{pmatrix} \vdots \\ a_{i1}(u_1 + v_1) + a_{i2}(u_2 + v_2) + \cdots + a_{in}(u_n + v_n) \\ \vdots \end{pmatrix} \\ = A \cdot \begin{pmatrix} u_1 + v_1 \\ u_2 + v_2 \\ \vdots \\ u_n + v_n \end{pmatrix} \\ = A \cdot (u + v).$$

The second statement is left as an exercise. $\square$

Proof. [Claim] $\Leftarrow$: Assume $b = 0$, we want to show that L is a subspace. Indeed $0 \in L$ because $A \cdot 0 = 0$.

If $x = (x_1, \dots, x_n), y = (y_1, \dots, y_n) \in L$, then

$$A \cdot (x + y) \stackrel{(1)}{=} A \cdot x + A \cdot y = 0 + 0 = 0.$$

Thus $x + y \in L$.

Let $x \in L, a \in K$. Then

$$A \cdot (a \cdot x) \stackrel{(2)}{=} a \cdot (A \cdot x) = a \cdot 0 = 0.$$

Thus $a \cdot x \in L$.

Together these three properties show that $L \subseteq K^n$ is a subspace.

$\Rightarrow$: Assume that L is a subspace, and we we'll show that $b = 0$. Indeed since $L \subseteq K^n$ is a subspace, $0_V \in L$. Thus $A \cdot 0_V = b$. But $A \cdot 0_V = 0_V$. Thus $b = 0_V$. $\square$

Example 3.13:

Let K be a field. Define

$$K^\infty := \{(a_1, a_2, \dots) \mid a_i \in K, i \in \mathbb{N}\}.$$

We turn K^∞ into a vector space as follows:

$$\begin{aligned} (a_1, a_2, \dots) + (b_1, b_2, \dots) &:= (a_1 + b_1, a_2 + b_2, \dots) \\ c \cdot (a_1, a_2, \dots) &:= (c \cdot a_1, c \cdot a_2, \dots) \end{aligned}$$

Recalling the first lecture, taking $K = \mathbb{R}$, we have

$$\text{Fib} := \{(a_1, a_2, \dots) \mid a_k = a_{k-1} + a_{k-2}, k \geq 3\}.$$

As an exercise, show that Fib is a subspace of $\mathbb{R}^\infty$. More generally, fix $\alpha, \beta \in \mathbb{R}$. Then

$$\mathcal{U}_{\alpha, \beta} := \{(a_1, a_2, \dots) \in \mathbb{R}^\infty \mid a_k = \alpha a_{k-1} + \beta a_{k-2}, k \geq 3\},$$

is a subspace of $\mathbb{R}^\infty$.

3.1 Spaces of Functions

Fix a field K and S is a non-empty set. Define

$$K^S := \{f : S \rightarrow K\}.$$

This denotes the set of all functions from S to K . We define the following operations on K^S :

$$\begin{aligned} (f + g)(s) &:= f(s) + g(s) \\ (a \cdot f)(s) &:= a \cdot (f(s)). \end{aligned}$$

With these operations, K^S is a vector space over K . (Exercise)

Exercise 3.14:

Let $S = [0, 1]$. Define $C(S) := \{f \in \mathbb{R}^S \mid f \text{ such that } f \text{ is continuous. Then } C(S) \text{ is a subspace of } \mathbb{R}^S.$

3.2 Polynomials

Let K be a field. Fix a letter called **FORMAL VARIABLE** x . A **POLYNOMIAL** over K (or polynomial with coefficients in K) is an expression of the form

$$f(x) = a_0 + a_1x + a_2x^2 + \cdots + a_nx^n.$$

where $n \in \mathbb{N}_0, a_i \in K \forall 0 \leq i \leq n$. We allow to omit 0-terms like $0 \cdot x^i$.

We call $f(x) = 0$ the **ZERO POLYNOMIAL**.

We can add and multiply polynomials as follows: Consider polynomials a and b we take $r = \max(n, m)$ and define

$$\begin{aligned} (a_0 + a_1x + \cdots + a_nx^n) + (b_0 + b_1x + \cdots + b_mx^m) &:= \\ (a_0 + b_0) + (a_1 + b_1)x + \cdots + (a_r + b_r)x^r. \end{aligned}$$

The multiplication is defined as

$$\begin{aligned} c \cdot (a_0 + a_1x + \cdots + a_nx^n) &:= \\ (c \cdot a_0) + (c \cdot a_1)x + \cdots + (c \cdot a_n)x^n. \end{aligned}$$

We write $K[x]$ for the set of all polynomials with coefficients in K .

Claim 3.15:

$K[x]$ is a vector space over K with the above operations.

Every $f(x) \in K[x]$ defines also a function $\tilde{f} : K \rightarrow K$. The subset of K^K which comes from polynomials is a subspace of K^K .

But: from $\tilde{f}$ we cannot always recover f .

Example 3.16:

Take $K = \mathcal{F}_2 = \{0, 1\}$. Consider $f(x) = x$ and $g(x) = x^2$. Then these are different polynomials, but $\tilde{f} = \tilde{g}$.

Definition 3.17: Degree of a polynomial

Let $f(x) = a_0 + a_1x + \dots + a_nx^n \in K[x]$ with $a_n \neq 0$. Then we define the **DEGREE** of f as $\deg(f) = n$. We define $\deg(0) = -\infty$.

Fix $d \in \mathbb{N}_0 \cup \{-\infty\}$. Define $K[x]_d$ to be all the polynomials of degree at most d . Then $K[x]_d$ is a subspace of $K[x]$.

Question 3.18:

Is $\{f(x) \in k[x] \mid \deg(f) = d\}$ also a subspace of $K[x]$?

Consider $\{p \in k[x]_5 \mid \sum_{n=0}^{\infty} a_n = 0\}$. Is this a subspace of $K[x]_5$?

Spaces of matrices: Consider $M_{m \times n}(K)$ to be the set of all $m \times n$ matrices with entries in K . Define $+$ and $\cdot$ as

$$(a_{ij}) + (b_{ij}) := (a_{ij} + b_{ij})$$

$$c \cdot (a_{ij}) := (c \cdot a_{ij}).$$

With these operations, $M_{m \times n}(K)$ is a vector space over K .

As a motivation for the next few lectures: Let V be a vector space, let $S \subseteq V$ be a subset. What is the smallest subspace of V containing S ?

4 Span

Consider the following lemma

Lemma 4.1:

Let V be a vector space and let $\{W_i\}_{i \in I}$ be a family of subspaces of V (i.e. $W_i \subseteq V$ is a subspace for all $i \in I$). Then the intersection

$$W := \bigcap_{i \in I} W_i$$

is a subspace of V .

Proof. Note that $0_V \in W_i$ for all $i \in I$, thus $0_V \in W$.

Let now $a, b \in K, v, w \in W$. We'll show that $a \cdot v + b \cdot w \in W$.

Let $j \in I$ be any index in I . Since $v, w \in W$, they must also be in W_j . Since W_j is a subspace of V , it follows that

$$a \cdot v + b \cdot w \in W_j.$$

But this holds $\forall j \in I$, therefore,

$$a \cdot v + b \cdot w \in \bigcap_{i \in I} W_i = W.$$

By a previous lemma, this shows that W is a subspace of V . $\square$

Let $S \subseteq V$ be a subset.

Definition 4.2: Span

Define the **SPAN** of S as

$$\text{Sp}(S) := \bigcap_{W \in \mathcal{N}} W,$$

where

$$\mathcal{N} := \{W \subseteq V \mid W \text{ is a subspace of } V, S \subseteq W\}.$$

Note that $\mathcal{N} \neq \emptyset$ since $V \in \mathcal{N}$.

By the previous lemma, $\text{Sp}(S)$ is a subspace of V . Sometimes, we also call it the subspace of V generated by S .

Lemma 4.3:

Among all subspaces of V that contain S , $\text{Sp}(S)$ is the smallest one, i.e.,

1. $\text{Sp}(S) \subseteq V$ is a subspace of V .
2. If $W \subseteq V$ is a subspace and $W \supseteq S$, then $W \supseteq \text{Sp}(S)$.

Proof. 1. This was shown above.

2. Let $W \subseteq V$ be a subspace with $W \supseteq S$. By definition of $\mathcal{N}$, we have $W \in \mathcal{N}$. Thus by definition of $\text{Sp}(S)$, we have

$$\text{Sp}(S) = \bigcap_{U \in \mathcal{N}} U \subseteq W.$$

$\square$

4.1 Linear Combinations

This definition is good from a theoretical point of view, but not very handy when we actually want to compute $\text{Sp}(S)$. We thus introduce the following concept.

Definition 4.4: Linear Combination

Let $n \in \mathbb{Z}_{\geq 1}$, $a_1, \dots, a_n \in K$ and $v_1, \dots, v_n \in V$. A **LINEAR COMBINATION** of the vectors $v_1, \dots, v_n$ with coefficients $a_1, \dots, a_n$ is the vector

$$a_1 \cdot v_1 + a_2 \cdot v_2 + \dots + a_n \cdot v_n.$$

We only look at finite linear combinations, i.e., n is always finite.

Lemma 4.5:

Let $\emptyset \neq S \subseteq V$. Then

$$\text{Sp}(S) = \{a_1 v_1 + \dots + a_n v_n \mid n \in \mathbb{Z}_{\geq 1}, v_i \in S, a_i \in K\}.$$

i.e. $\text{Sp}(S)$ is the subset of V obtained by taking all possible linear combinations of vectors in S .

Proof. Denote $\widetilde{\text{Sp}}(S) := \{a_1 v_1 + \dots + a_n v_n \mid n \in \mathbb{Z}_{\geq 1}, v_i \in S, a_i \in K\}$. We will show that $\widetilde{\text{Sp}}(S) = \text{Sp}(S)$.

We first show that 1) $S \subseteq \widetilde{\text{Sp}}(S)$, 2) $\widetilde{\text{Sp}}(S)$ is a subspace of V and 3) $\forall$ subspace $W \subseteq V$ with $S \subseteq W$ we have $\widetilde{\text{Sp}}(S) \subseteq W$.

Note that this is useful since 1)+2) would imply that $\text{Sp}(S) \subseteq \widetilde{\text{Sp}}(S)$ by the previous lemma, and 3) would imply $\widetilde{\text{Sp}}(S) \subseteq \bigcap_{W \in \mathcal{N}} W = \text{Sp}(S)$. Together, this implies $\widetilde{\text{Sp}}(S) = \text{Sp}(S)$.

It remains to prove 1), 2) and 3).

1) If $v \in S$, then $v = 1 \cdot v \in \widetilde{\text{Sp}}(S)$.

2) We'll show that $0_V \in \widetilde{\text{Sp}}(S)$ and that $\forall \alpha, \beta \in K, \forall v, w \in \widetilde{\text{Sp}}(S) : \alpha \cdot v + \beta \cdot w \in \widetilde{\text{Sp}}(S)$.

Clearly, $0_V = 0 \cdot v \in \widetilde{\text{Sp}}(S)$ for any $v \in S$.

Write $v = a_1 v_1 + \dots + a_n v_n$ and $w = b_1 w_1 + \dots + b_m w_m$. Then

$$\begin{aligned} \alpha \cdot v + \beta \cdot w &= \alpha(a_1 v_1 + \dots + a_n v_n) + \beta(b_1 w_1 + \dots + b_m w_m) \\ &= (\alpha a_1) v_1 + \dots + (\alpha a_n) v_n + (\beta b_1) w_1 + \dots + (\beta b_m) w_m. \end{aligned}$$

This is a linear combination of vectors in S , thus $\alpha \cdot v + \beta \cdot w \in \widetilde{\text{Sp}}(S)$.

3) Let $W \subseteq V$ be a subspace with $S \subseteq W$. Let v be in $\widetilde{\text{Sp}}(S)$. We need to prove that $v \in W$. Indeed, write $v = a_1 v_1 + \dots + a_n v_n$ with $v_i \in S$. By assumption, $S \subseteq W$, thus $v_i \in W$. As W is a linear subspace, it follows that $v \in W$. $\square$

We have shown that the span of a subset S of a vector space V can be written in two ways. The first one is more theoretical, the second one more practical.

The span of the empty set is $\{0_V\}$. From now on, let's agree that linear combinations of elements of the empty set is just 0_V .

As a notation: Many times $S = \{v_1, \dots, v_n\}$ is a finite set. We write $\text{Sp}\{v_1, \dots, v_n\}$ or $\text{Sp}(v_1, \dots, v_n)$ instead of $\text{Sp}(\{v_1, \dots, v_n\})$.

Definition 4.6: Generating/Spanning Set

Let V be a vector space and $S \subseteq V$ a subset. We say that S **GENERATES** or **SPANS** V if $\text{Sp}(S) = V$.

If $W \subseteq V$ is a subspace, we say that S generates or spans W if $\text{Sp}(S) = W$.

Definition 4.7: Finite Dimensional Vector Space

A vector space V is **FINITE DIMENSIONAL** if there exists a finite set $S \subseteq V$ that generates V .

If such a finite S does not exist, then V is called **INFINITE DIMENSIONAL**.

Example 4.8: Subspaces of $\mathbb{R}^2$

Let $V = \mathbb{R}^2, K = \mathbb{R}$. Some subspaces of V are:

1. $\{0_{\mathbb{R}^2}\} = \{(0, 0)\} = \text{Sp}(\emptyset) = \text{Sp}\{0_{\mathbb{R}^2}\}$
2. Let $0 \neq v = (a, b) \in \mathbb{R}^2$. Then

$$\text{Sp}\{v\} = \{\alpha \cdot (a, b) \mid \alpha \in \mathbb{R}\} = \{(\alpha a, \alpha b) \mid \alpha \in \mathbb{R}\}.$$

Geometrically speaking, this is the line through the origin and v .

3. Let $0 \neq v_1, w \in \mathbb{R}^2$. If $\text{Sp}\{v\} = \text{Sp}\{w\}$, this is equivalent to saying that $\exists \alpha \in \mathbb{R} \setminus \{0\}$ such that $w = \alpha v$.

4. Let $W \subseteq \mathbb{R}^2$ be a subspace and assume that $W = \text{Sp}\{v, w\}$ where $v \neq 0, w \notin \text{Sp}\{v\}$. Then $W = \mathbb{R}^2$.

Together this tells us that there are only three types of subspaces of $\mathbb{R}^2$: the zero subspace, lines through the origin, and the whole space $\mathbb{R}^2$.

Question: Let $v_1, \dots, v_n \in V$ and $w \in K^m$. How can we determine whether or not $w \in \text{Sp}\{v_1, \dots, v_n\}$?

e.g. let $v_1 = (1, 3), v_2 = (7, 73)$ and $w = (11, 137)$. In this case yes, $w \in \text{Sp}\{v_1, v_2\}$ as $-3 \cdot v_1 + 2 \cdot v_2 = w$.

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Solution: Take the column vectors $v_1, \dots, v_n$ and make a $m \times n$ matrix out of them:

$$A = \begin{pmatrix} | & | & \dots & | \\ v_1 & v_2 & & v_n \\ | & | & & | \end{pmatrix}.$$

If $x = (x_1, \dots, x_n) \in K^n$, then

$$A \cdot x = x_1 v_1 + x_2 v_2 + \dots + x_n v_n = \sum_{i=1}^n x_i v_i.$$

Given $w \in K^m$, the question is whether or not the system of equations $A \cdot x = w$ has a solution. If a solution exists, then $w \in \text{Sp}\{v_1, \dots, v_n\}$. If a solution does not exist, then $w \notin \text{Sp}\{v_1, \dots, v_n\}$.

In the first case, any solution x will give us a choice of coefficients $x_1, \dots, x_n$ such that $w = \sum_{i=1}^n x_i v_i$.

Example 4.9:

1) Let $V = K^n$. Denote by $e_1 := (1, 0, 0, \dots, 0)$, $e_2 := (0, 1, 0, \dots, 0)$, $\dots$, $e_n := (0, 0, 0, \dots, 1)$. Then

$$\text{Sp}\{e_1, e_2, \dots, e_n\} = K^n.$$

2) Let $W = k[x]_d$ (polynomials in x of degree at most d). Then

$$W = \text{Sp}\{1, x, x^2, \dots, x^d\}.$$

Similarly, take $V = K[x]$. Then

$$V = \text{Sp}\{1, x, x^2, x^3, \dots\} \text{ (infinite set) .}$$

3) Let $M = M_{m \times n}(K)$ be the set of all $m \times n$ matrices. For every $1 \leq i \leq m, 1 \leq j \leq n$, define the matrix

$$E_{ij} := \begin{pmatrix} 0 & \dots & 0 & \dots & 0 \\ \vdots & & \vdots & & \vdots \\ 0 & \dots & 1 & \dots & 0 \\ \vdots & & \vdots & & \vdots \\ 0 & \dots & 0 & \dots & 0 \end{pmatrix}.$$

i.e. the matrix with a 1 in the (i, j) -th position and 0 elsewhere.

Then

$$M = \text{Sp}\{E_{ij} \mid 1 \leq i \leq m, 1 \leq j \leq n\}.$$

All of these examples are finite dimensional vector spaces except $K[x]$. As an exercise, show that $K[x]$ is infinite dimensional.

5 Linear Independence

Let V be a vector space over K

Definition 5.1:

Let $v_1, \dots, v_n$ be a list of n vectors in V . We say that $v_1, \dots, v_n$ are **LINEARLY INDEPENDENT** if the only linear combination of $v_1, \dots, v_n$ that equals 0_V is the one with all coefficients equal to zero.

If the vectors are not linearly independent, they are called **LINEARLY DEPENDENT**.

Alternatively, $v_1, \dots, v_n$ are linearly independent iff

$\forall a_1, \dots, a_n \in K$ for which $a_1 v_1 + \dots + a_n v_n = 0_V$, we must have $a_1 = a_2 = \dots = a_n = 0$.

Let's agree that $\emptyset$ is linearly independent.

Remark 5.2:

1) 0_V can always be written as a linear combination of any list of vectors as $0_V = 0 \cdot v_1 + 0 \cdot v_2 + \dots + 0 \cdot v_n$. This is called the **TRIVIAL** linear combination.

2) Suppose that in our list there is a repetition, say $v_i = v_j$ for some $i \neq j$. Then the list is linearly dependent, since

$$0_V = 1 \cdot v_i + (-1) \cdot v_j + 0 \cdot v_1 + \dots + 0 \cdot v_n.$$

3) The order of the elements in the list does not matter.

4) If 0 is one of the elements in the list, then this list cannot be linearly independent.

As a generalization for families of vectors consider the following definition.

Definition 5.3:

Let $\mathcal{F} = \{v_i\}_{i \in \mathcal{I}}$ be a family of vectors in V . We say that $\mathcal{F}$ is **LINEARLY INDEPENDENT** if $\forall n \in \mathbb{Z}_{\geq 1}$ and any sequence of distinct indices $i_1, \dots, i_n \in \mathcal{I}$, the list of vectors $v_{i_1}, \dots, v_{i_n}$ is linearly independent.

Let $\emptyset \neq S \subseteq V$ be a subset. We say that S is linearly independent if every finite list of distinct vectors in S is linearly independent.

Lemma 5.4:

If $\emptyset \neq S \subseteq V$, is linearly independent, then every subset of it is also linearly independent.

Proof. Assume $\exists a_1, \dots, a_n \in K$ and $v_1, \dots, v_n \in W \subseteq S$ such that $a_1 v_1 + \dots + a_n v_n = 0_V$ with not all a_i equal to zero. Since $W \subseteq S$, this contradicts the linear independence of S . $\square$

Example 5.5:

1) $e_1, \dots, e_n \in K^n$ are linearly independent.

Solution. Suppose $a_1 e_1 + \dots + a_n e_n = 0$. This means that

$$a_1 \begin{pmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix} + a_2 \begin{pmatrix} 0 \\ 1 \\ \vdots \\ 0 \end{pmatrix} + \dots + a_n \begin{pmatrix} 0 \\ 0 \\ \vdots \\ 1 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ \vdots \\ 0 \end{pmatrix}.$$

But since we are working component-wise, this means that

$$\begin{pmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ \vdots \\ 0 \end{pmatrix}.$$

In particular, $a_i = 0$ for all $1 \leq i \leq n$.

2) The set $\{e_1, e_2, \dots\} \subseteq K^\infty$ is linearly independent.

3) The set $\{1, x, x^2, x^3, \dots, x^d\} \subseteq K[x]_d$ is linearly independent.

4) The set $\{1, x, x^2, x^3, \dots\} \subseteq K[x]$ is linearly independent.

5) Let V be a vector space, let $v \in V$. Show that $\{v\}$ is linearly independent iff $v \neq 0_V$.

6 Basis

The following is one of the most central definitions in linear algebra.

Definition 6.1: Basis

A subset $S \subseteq V$ is called a **basis** of V if the following holds:

- 1) S is linearly independent.
- 2) S spans V , i.e. $\text{Sp}(S) = V$.

Proposition 6.2:

A subset $S \subseteq V$ is a basis of V iff every $v \in V$ can be written in a *unique* way as a linear combination of vectors from S .

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The meaning of "unique" in the above proposition is the following: suppose $v \in V$ is written as $v =$ linear combination #1 with vectors from S , and also as $v =$ linear combination #2 with vectors from S .

Define C_1 to be all the vectors that appear in linear combination #1 without zero coefficient. Similarly, define C_2 .

Then $C_1 = C_2$ and $\forall u \in C_1$, the coefficient of u in #1 equals the coefficient of u in #2.

In case, S is a finite set, say $S = \{u_1, \dots, u_n\}$, then uniqueness means that if $v = a_1 u_1 + \dots + a_n u_n = b_1 u_1 + \dots + b_n u_n$, then $a_i = b_i$ for all $1 \leq i \leq n$.

Proof. [of proposition] For simplicity assume that S is finite. (The general case is an exercise.)

$\Leftarrow$ Assume every $v \in V$ can be written in a unique way. Because every $v \in V$ can be written, this implies that $\text{Sp}(S) = V$. We'll show now that S is linearly independent.

Write $S = \{v_1, \dots, v_n\}$. Consider the 0_V . Clearly, $0 = 0 \cdot v_1 + \dots + 0 \cdot v_n$. But by assumption, there is a unique linear combination of $v_1, \dots, v_n$ that equals 0_V . It follows that if

$$a_1 v_1 + \dots + a_n v_n = 0_V.$$

Then all a_i must be zero. Thus, S is linearly independent. Thus, S is a basis of V .

$\Rightarrow$ Assume S is a basis of V . Therefore, $\text{Sp}(S) = V$. This implies that every $v \in V$ can be written as a linear combination of vectors from S . It remains to show uniqueness.

Assume by contradiction that $\exists v \in V$ can be written as

$$v = a_1 v_1 + \dots + a_n v_n = b_1 v_1 + \dots + b_n v_n.$$

Where $a_k \neq b_k$ for some $1 \leq k \leq n$.

Now, subtracting both expressions, we get

$$0_V = (a_1 - b_1)v_1 + \dots + \underbrace{(a_k - b_k)}_{\neq 0} v_k + \dots + (a_n - b_n)v_n.$$

Thus 0_V can be written as a non-trivial linear combination of vectors from S . Thus S is not linearly independent, which contradicts our assumption that S is a basis. $\square$

Example 6.3:

- 1) The set $\{e_1, \dots, e_n\}$ is a basis of K^n .
- 2) The set $\{1, x, x^2, \dots, x^d\}$ is a basis of $K[x]_d$.
- 3) The set $\{E_{ij} \mid 1 \leq i \leq m, 1 \leq j \leq n\}$ is a basis of $M_{m \times n}(K)$.

In case of K^n , we call the basis $\{e_1, \dots, e_n\}$ the **STANDARD BASIS** or **CANONICAL BASIS** of K^n .

Lemma 6.4:

Let $v_1, \dots, v_m \in V$ be a list of linearly dependent vectors. Then $\exists 1 \leq j \leq m$ such that

- 1) $v_j \in \text{Sp}(v_1, \dots, v_{j-1})$
- 2) $\text{Sp}(v_1, \dots, v_m) = \text{Sp}(v_1, \dots, v_{j-1}, v_{j+1}, \dots, v_m)$

Proof. Since the list is linearly dependent, $\exists a_1, \dots, a_m \in K$ not all of them zero, such that

$$a_1 v_1 + \dots + a_m v_m = 0_V.$$

Define $j := \max\{i \mid a_i \neq 0\}$ (i.e. $a_j \neq 0$ and $a_{j+1} = \dots = a_m = 0$).

Then $a_1 v_1 + \dots + a_{j-1} v_{j-1} + a_j v_j = 0_V$. Since $a_j \neq 0$, we can solve for v_j :

$$v_j = -\frac{a_1}{a_j} v_1 - \dots - \frac{a_{j-1}}{a_j} v_{j-1}. \quad (6.1)$$

This implies $v_j \in \text{Sp}(v_1, \dots, v_{j-1})$.

To prove 2), let $v \in \text{Sp}(v_1, \dots, v_m)$. Then

$$v = c_1 v_1 + \dots + c_m v_m.$$

Substituting (6.1) into this expression, we obtain, that v is a linear combination of $v_1, \dots, v_{j-1}, v_{j+1}, \dots, v_m$. Hence,

$$\text{Sp}(v_1, \dots, v_m) \subseteq \text{Sp}(v_1, \dots, v_{j-1}, v_{j+1}, \dots, v_m).$$

But clearly, we also have the other inclusion, thus

$$\text{Sp}(v_1, \dots, v_m) = \text{Sp}(v_1, \dots, v_{j-1}, v_{j+1}, \dots, v_m).$$

□

Lemma 6.5:

Let $v_1, \dots, v_m$ be a list of linearly dependent vectors such that $v_1, \dots, v_k$ are linearly independent for some $1 \leq k < m$. Then j from Lemma 6.4 satisfies $k < j$.

Proof. Assume by contradiction that $j \leq k$. We have:

$$v_j = a_1 v_1 + \dots + a_{j-1} v_{j-1},$$

for some $a_1, \dots, a_{j-1} \in K$.

But this implies that

$$0 = a_1 v_1 + \dots + a_{j-1} v_{j-1} - 1 \cdot v_j.$$

Since there is a -1 coefficient, this is a non-trivial linear combination of $v_1, \dots, v_k$ that equals 0_V . This contradicts the linear independence of $v_1, \dots, v_k$. □

Lemma 6.6:

Let $w_1, \dots, w_n \in V$ such that $\text{Sp}(w_1, \dots, w_n) = V$ and let $v \in V$. Then the list $v, w_1, \dots, w_n$ is linearly dependent.

Proof. Write $v = a_1 w_1 + \dots + a_n w_n$ for some $a_1, \dots, a_n \in K$. Then

$$0_V = (-1) \cdot v + a_1 w_1 + \dots + a_n w_n.$$

This is a non-trivial linear combination of $v, w_1, \dots, w_n$ that equals 0_V . Thus, the list is linearly dependent. □

Lemma 6.7:

Let $v_1, \dots, v_n \in V$ such that $\text{Sp}(v_1, \dots, v_n) = V$. Let $u_1, \dots, u_m$ be a list of linearly independent vectors in V . Then $m \leq n$.

Proof. The proof will have m steps.

Step 1: We will replace u_1 by one of the vectors $v_1, \dots, v_n$. How? Consider the list $u_1, v_1, \dots, v_n$. By Lemma 6.6, this list is linearly dependent. By Lemma 6.4, one of the vectors in this list is in the span of the vectors that appear before it, and if we drop this vector from the list, the overall span does not change. By Lemma 6.5, this vector cannot be u_1 (because u_1 is linearly independent).

We now drop that vector and get a list of n vectors that still span V and the first vector is u_1 .

Step j ($2 \leq j \leq m$): From step $j-1$, we should have a list of n vectors that span V and the first $j-1$ vectors are $u_1, \dots, u_{j-1}$ and the other rest $n - (j-1)$ of vectors are taken from the list $v_1, \dots, v_n$.

Let's write this list of n vectors as $u_1, \dots, u_{j-1}, w_1, \dots, w_{n-(j-1)}$. Consider now the list $u_1, \dots, u_{j-1}, u_j, w_1, \dots, w_{n-(j-1)}$. □